

AHC RO/ARO — COMPUTER COMPLETE ONE-LINER CAPSULE

1. Computer Fundamentals

1. A computer is an electronic programmable device that accepts data, processes it and produces information.
 2. The basic working cycle of a computer is **Input** → **Processing** → **Output** → **Storage**.
 3. Raw facts and figures are called **Data**.
 4. Processed and meaningful data is called **Information**.
 5. The instructions given to a computer are called **Commands/Instructions**.
 6. A collection of instructions written to perform a task is called a **Program**.
 7. The physical components of a computer are called **Hardware**.
 8. A set of programs/instructions is called **Software**.
 9. **GIGO** stands for Garbage In, Garbage Out.
 10. GIGO means incorrect input generally produces incorrect output.
 11. The main characteristics of a computer are speed, accuracy, storage, diligence, automation and versatility.
 12. A computer does not possess human-like common sense by itself.
 13. The basic unit of digital information is the **bit**.
 14. A bit can represent **0 or 1**.
 15. A group of 4 bits is called a **Nibble**.
 16. A group of 8 bits is called a **Byte**.
 17. A computer represents information internally mainly using the **binary number system**.
 18. The two basic binary digits are **0 and 1**.
 19. The central processing unit is called the **CPU**.
 20. CPU is often called the **brain of the computer**, although it is not literally the whole computer.
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2. CPU

21. CPU stands for **Central Processing Unit**.
 22. The major functional components of CPU include **ALU, Control Unit and Registers**.
 23. ALU stands for **Arithmetic Logic Unit**.
 24. ALU performs arithmetic and logical operations.
 25. Addition, subtraction, multiplication and division are arithmetic operations.
 26. Comparisons and logical operations are handled by the ALU.
 27. CU stands for **Control Unit**.
 28. The Control Unit controls and coordinates the activities of the computer.
 29. Registers are very high-speed storage locations inside/associated closely with the CPU.
 30. Registers are faster than main memory.
 31. The **Program Counter (PC)** holds the address of the next instruction to be executed.
 32. The **Instruction Register (IR)** holds the current instruction being executed.
 33. The **Accumulator** stores intermediate arithmetic/logic results in many CPU architectures.
 34. The **MAR** holds the address of the memory location to be accessed.
 35. The **MDR/MBR** holds data being transferred to/from memory.
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3. Instruction Cycle

- 36. The basic instruction cycle includes **Fetch** → **Decode** → **Execute**.
 - 37. During the fetch phase, the instruction is obtained from memory.
 - 38. During decode, the CPU interprets the instruction.
 - 39. During execution, the required operation is performed.
 - 40. The **Program Counter** helps identify the next instruction.
 - 41. The clock provides timing signals for CPU operations.
 - 42. CPU performance is commonly associated with clock frequency, but clock speed alone does not determine total performance.
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4. Hardware

- 43. Hardware means the physical/tangible components of a computer.
 - 44. Keyboard is an **input device**.
 - 45. Monitor is an **output device**.
 - 46. Printer is an **output device**.
 - 47. Scanner is an **input device**.
 - 48. Touchscreen can function as both an input and output device.
 - 49. A microphone is an input device.
 - 50. A speaker is an output device.
 - 51. A webcam is an input device.
 - 52. A projector is an output device.
 - 53. A plotter is mainly used for high-quality large drawings/graphics.
 - 54. A joystick is primarily an input device.
 - 55. A light pen is an input device.
 - 56. A barcode reader is an input device.
 - 57. A QR-code scanner is an input device.
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5. OCR, OMR, MICR

- 58. **OCR** stands for Optical Character Recognition.
 - 59. OCR converts recognized printed/handwritten characters into machine-readable text.
 - 60. **OMR** stands for Optical Mark Recognition.
 - 61. OMR detects marked responses on specially designed forms.
 - 62. OMR is commonly associated with objective answer sheets.
 - 63. **MICR** stands for Magnetic Ink Character Recognition.
 - 64. MICR is widely used in banking for cheque processing.
 - 65. MICR characters are printed using magnetizable ink.
 - 66. OCR reads characters optically, whereas MICR uses magnetic characteristics of special ink.
 - 67. OMR detects marks rather than interpreting normal text characters.
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6. Printers

- 68. Printers are broadly classified into **Impact and Non-Impact printers**.
- 69. Dot Matrix is an **impact printer**.
- 70. Daisy Wheel is an impact printer.
- 71. Line Printer is an impact printer.
- 72. Laser Printer is a **non-impact printer**.

- 73. Inkjet Printer is a non-impact printer.
 - 74. Thermal printers are non-impact printers.
 - 75. Impact printers physically strike the printing surface/ribbon.
 - 76. Laser printers generally use toner.
 - 77. Inkjet printers use liquid ink.
 - 78. Dot Matrix printers are suitable where multipart forms/carbon copies may be required.
 - 79. Laser printers generally provide fast, high-quality text printing.
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7. Memory — VERY IMPORTANT

- 80. Computer memory is broadly divided into **Primary and Secondary Memory**.
 - 81. Primary memory is directly accessible by the CPU.
 - 82. RAM is a primary memory.
 - 83. ROM is a primary memory.
 - 84. Cache memory is high-speed memory located close to the CPU.
 - 85. Registers are among the fastest storage locations in a computer.
 - 86. Secondary storage is used for long-term data storage.
 - 87. Hard Disk and SSD are secondary storage devices.
 - 88. RAM stands for **Random Access Memory**.
 - 89. RAM is generally **volatile memory**.
 - 90. Data stored in RAM is generally lost when power is removed.
 - 91. ROM stands for **Read Only Memory**.
 - 92. ROM is **non-volatile**.
 - 93. Data in ROM is retained even after power is switched off.
 - 94. SRAM stands for Static RAM.
 - 95. DRAM stands for Dynamic RAM.
 - 96. SRAM is faster and more expensive than DRAM.
 - 97. DRAM is commonly used as main memory.
 - 98. Cache is generally faster than RAM.
 - 99. Cache reduces the average time required by the CPU to access frequently used data/instructions.
 - 100. The general speed hierarchy is **Registers → Cache → RAM → Secondary Storage**.
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8. ROM TYPES

- 101. PROM stands for **Programmable Read Only Memory**.
 - 102. PROM can generally be programmed once.
 - 103. EPROM stands for **Erasable Programmable Read Only Memory**.
 - 104. EPROM can be erased using ultraviolet light.
 - 105. EEPROM stands for **Electrically Erasable Programmable Read Only Memory**.
 - 106. EEPROM can be electrically erased and reprogrammed.
 - 107. Flash memory is a type of electrically erasable non-volatile memory technology.
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9. Memory Units

- 108. 1 Byte = **8 bits**.
- 109. 1 Nibble = **4 bits**.
- 110. Traditionally, 1 KB = **1024 bytes** in binary-based exam conventions.

- 111. 1 MB = 1024 KB.
- 112. 1 GB = 1024 MB.
- 113. 1 TB = 1024 GB.
- 114. 1 PB = 1024 TB.
- 115. Order: **Bit** → **Byte** → **KB** → **MB** → **GB** → **TB** → **PB**.

Quick trick

B K M G T P

10. Secondary Storage

- 116. Hard Disk Drive is magnetic storage.
 - 117. SSD stands for **Solid State Drive**.
 - 118. SSD has no moving mechanical disk platter.
 - 119. SSD is generally faster than a traditional HDD.
 - 120. Pen Drive is a flash-memory-based storage device.
 - 121. CD stands for Compact Disc.
 - 122. DVD stands for Digital Versatile Disc.
 - 123. Blu-ray provides higher storage capacity than conventional DVD.
 - 124. Memory cards generally use flash memory.
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11. Number System

- 125. Binary number system has base **2**.
 - 126. Octal number system has base **8**.
 - 127. Decimal number system has base **10**.
 - 128. Hexadecimal number system has base **16**.
 - 129. Binary digits are **0 and 1**.
 - 130. Octal digits range from **0 to 7**.
 - 131. Decimal digits range from **0 to 9**.
 - 132. Hexadecimal uses **0–9 and A–F**.
 - 133. In hexadecimal, A=10, B=11, C=12, D=13, E=14 and F=15.
 - 134. Binary 1010 equals decimal **10**.
 - 135. Binary 1111 equals decimal **15**.
 - 136. Decimal 8 equals binary **1000**.
 - 137. Decimal 15 equals binary **1111**.
 - 138. 1 hexadecimal digit represents **4 binary bits**.
 - 139. 1 octal digit represents **3 binary bits**.
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12. Complements

- 140. 1's complement is obtained by changing **0→1 and 1→0**.
 - 141. 2's complement is obtained by taking the 1's complement and adding 1.
 - 142. 2's complement is widely used for representing signed integers in digital computers.
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13. BCD

- 143. BCD stands for **Binary Coded Decimal**.
 - 144. In BCD, each decimal digit is represented separately using four bits.
 - 145. Decimal 59 in BCD is **0101 1001**.
 - 146. BCD should not be confused with ordinary binary representation of the whole number.
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14. ASCII & Unicode

- 147. ASCII stands for **American Standard Code for Information Interchange**.
 - 148. Traditional ASCII uses **7 bits**.
 - 149. Extended ASCII is commonly associated with 8-bit representations.
 - 150. Unicode is designed to represent characters from many writing systems.
 - 151. Unicode is much broader than traditional ASCII.
 - 152. UTF-8 is a widely used Unicode encoding.
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15. Software

- 153. Software is a set of programs/instructions.
 - 154. Software is broadly classified as **System Software and Application Software**.
 - 155. Operating System is system software.
 - 156. Device drivers are system software.
 - 157. Utility programs are generally categorized as system software.
 - 158. MS Word is application software.
 - 159. MS Excel is application software.
 - 160. PowerPoint is application software.
 - 161. A compiler is system software/programming software used to translate source code.
 - 162. Antivirus software is utility/security software.
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16. Operating System

- 163. An Operating System acts as an interface between the user/application programs and computer hardware.
 - 164. Windows, Linux, macOS and Android are operating systems.
 - 165. OS performs process management.
 - 166. OS performs memory management.
 - 167. OS performs file management.
 - 168. OS manages input/output devices.
 - 169. OS provides security and access control mechanisms.
 - 170. OS allocates system resources among programs/processes.
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17. Types of Operating Systems

- 171. Single-user OS is designed primarily for one user at a time.
 - 172. Multi-user OS supports multiple users.
 - 173. Multitasking OS allows multiple tasks/processes to be handled concurrently.
 - 174. Multiprocessing uses more than one processor/core.
 - 175. Real-time OS is designed to respond within specified timing constraints.
 - 176. Distributed OS coordinates resources across distributed systems.
 - 177. Network OS provides network-related management/services.
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18. Booting

- 178. Booting is the process of starting/restarting a computer and loading the operating system.
 - 179. Cold boot means starting from a powered-off state.
 - 180. Warm boot means restarting an already running system.
 - 181. BIOS stands for **Basic Input/Output System**.
 - 182. UEFI is a modern firmware interface that largely replaces traditional BIOS functionality.
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19. GUI / CLI

- 183. GUI stands for **Graphical User Interface**.
 - 184. CLI stands for **Command Line Interface**.
 - 185. GUI uses graphical elements such as windows, icons and menus.
 - 186. CLI primarily uses typed commands.
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20. Computer Generations

- 187. First generation computers used **Vacuum Tubes**.
- 188. Second generation used **Transistors**.
- 189. Third generation used **Integrated Circuits**.
- 190. Fourth generation used **Microprocessors**.
- 191. Fifth generation is associated with AI/advanced intelligent computing concepts.

Memory trick

Tube → Transistor → IC → Microprocessor → AI

21. Computer Languages

- 192. Machine language consists of binary machine instructions.
- 193. Assembly language uses mnemonic codes.
- 194. High-level languages are closer to human-readable syntax.
- 195. C, C++, Java and Python are high-level programming languages.
- 196. Assembly language generally requires an **assembler**.
- 197. High-level language translation may involve a compiler or interpreter.

22. Compiler / Interpreter / Assembler

- 198. A compiler translates source code into target/object code.
- 199. An interpreter generally translates and executes program instructions incrementally.
- 200. An assembler converts assembly language into machine/object code.
- 201. A compiler typically reports errors after analyzing/compiling relevant portions rather than executing source directly.
- 202. Syntax errors arise when language grammar/rules are violated.
- 203. Logical errors may allow a program to run but produce incorrect results.
- 204. Runtime errors occur during program execution.

23. Algorithm

- 205. An algorithm is a finite, ordered, step-by-step procedure for solving a problem.
- 206. An algorithm should have clearly defined steps.
- 207. A flowchart is a graphical representation of an algorithm/process.
- 208. A decision in a flowchart is generally represented by a **diamond**.
- 209. Process is generally represented by a **rectangle**.
- 210. Start/End is generally represented by an **oval/terminator**.
- 211. Input/Output is generally represented by a **parallelogram**.

24. Computer Networks

- 212. A computer network connects devices for communication/resource sharing.
- 213. PAN stands for **Personal Area Network**.
- 214. LAN stands for **Local Area Network**.
- 215. MAN stands for **Metropolitan Area Network**.
- 216. WAN stands for **Wide Area Network**.
- 217. LAN generally covers a small geographical area.
- 218. MAN generally covers a city/metropolitan area.
- 219. WAN covers large geographical areas.
- 220. The Internet is the world's largest interconnected network of networks.

25. Network Topology

- 221. Bus topology uses a common backbone.
 - 222. Star topology uses a central connecting device.
 - 223. Ring topology connects nodes in a circular arrangement.
 - 224. Mesh topology provides multiple interconnections among nodes.
 - 225. Tree topology has a hierarchical structure.
 - 226. Hybrid topology combines characteristics of multiple topologies.
 - 227. Star topology is widely used in modern Ethernet LANs.
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26. Networking Devices

- 228. Hub broadcasts incoming data to connected ports.
 - 229. Switch forwards frames using MAC-address information.
 - 230. Router forwards packets between different networks.
 - 231. Repeater regenerates/strengthens signals.
 - 232. Bridge connects network segments at the data-link layer.
 - 233. Gateway can connect networks using different protocols/architectures.
 - 234. Modem converts between digital and analog signaling forms for suitable communication links.
 - 235. Modem = **Modulator** + **Demodulator**.
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27. MAC & IP

- 236. MAC stands for **Media Access Control**.
 - 237. A MAC address is associated with a network interface.
 - 238. IP stands for **Internet Protocol**.
 - 239. IP address is a logical network address.
 - 240. IPv4 uses **32-bit addresses**.
 - 241. IPv6 uses **128-bit addresses**.
 - 242. IPv6 provides a vastly larger address space than IPv4.
 - 243. MAC and IP addresses serve different purposes and operate at different networking layers.
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28. Important Protocols

- 244. HTTP = HyperText Transfer Protocol.
- 245. HTTPS = HyperText Transfer Protocol Secure.
- 246. FTP = File Transfer Protocol.
- 247. SMTP = Simple Mail Transfer Protocol.
- 248. POP3 = Post Office Protocol version 3.
- 249. IMAP = Internet Message Access Protocol.
- 250. DNS = Domain Name System.
- 251. DHCP = Dynamic Host Configuration Protocol.
- 252. TCP = Transmission Control Protocol.
- 253. UDP = User Datagram Protocol.
- 254. IP = Internet Protocol.

Important

SMTP → sending email

POP3/IMAP → receiving/accessing email

FTP → file transfer

DNS → domain name resolution

DHCP → automatic network configuration

29. Internet

- 255. Internet is a global network of interconnected networks.
 - 256. WWW stands for **World Wide Web**.
 - 257. WWW is a service that operates over the Internet.
 - 258. Internet and WWW are **not synonymous**.
 - 259. A website is a collection of related web resources/pages.
 - 260. A webpage is an individual web document/resource.
 - 261. A web browser is software used to access web content.
 - 262. Chrome, Firefox and Edge are browsers.
 - 263. A search engine helps locate information on the web.
 - 264. Google and Bing are search engines.
 - 265. Browser and search engine are not the same thing.
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30. URL / Domain

- 266. URL stands for **Uniform Resource Locator**.
 - 267. URL identifies the location/address of a web resource.
 - 268. A domain name is a human-readable name associated with an Internet resource.
 - 269. DNS translates/resolves domain names to IP addresses.
 - 270. .com is a commonly used generic top-level domain.
 - 271. .org is traditionally associated with organizations.
 - 272. .gov is commonly associated with government entities in relevant national domain systems.
 - 273. .in is India's country-code top-level domain.
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31. Email

- 274. Email means electronic mail.
 - 275. SMTP is primarily used for sending mail.
 - 276. POP3 downloads/accesses mail from a server.
 - 277. IMAP synchronizes/accesses messages while retaining server-side organization.
 - 278. CC means **Carbon Copy**.
 - 279. BCC means **Blind Carbon Copy**.
 - 280. Recipients in BCC are hidden from other recipients.
 - 281. An email attachment is a file sent along with an email message.
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32. Cyber Security

- 282. Cybersecurity protects computer systems, networks and data from unauthorized/malicious activity.
- 283. Malware means malicious software.
- 284. Virus is a malicious program that typically attaches to a host file/program.
- 285. Worm can self-replicate and spread across networks without requiring a host file in the same way as a virus.
- 286. Trojan disguises itself as legitimate/useful software.
- 287. Ransomware can encrypt/lock data and demand payment.

- 288. Spyware secretly monitors/collects user activity.
 - 289. Adware displays unwanted advertisements and may track activity depending on the software.
 - 290. Phishing uses deceptive messages/sites to steal information.
 - 291. Smishing is phishing through SMS/text messages.
 - 292. Vishing is phishing through voice calls.
 - 293. Pharming redirects users to fraudulent websites.
 - 294. Spoofing involves falsifying identity/source information.
 - 295. Firewall filters/controls network traffic according to security rules.
 - 296. Antivirus software detects/prevents/removes many forms of malware.
 - 297. Strong passwords should be unique and difficult to guess.
 - 298. Multi-factor authentication adds an additional authentication factor beyond a password.
 - 299. Encryption converts readable plaintext into protected ciphertext using an encryption process.
 - 300. Decryption converts ciphertext back into readable plaintext.
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33. Cloud Computing

- 301. Cloud computing provides computing resources/services over a network, usually the Internet.
- 302. IaaS = Infrastructure as a Service.
- 303. PaaS = Platform as a Service.
- 304. SaaS = Software as a Service.
- 305. Google Drive is an example of cloud-based storage/service.
- 306. AWS is a major cloud platform/provider.
- 307. Microsoft Azure is a cloud platform.
- 308. SaaS provides ready-to-use software applications.
- 309. PaaS provides a platform/environment for application development and deployment.
- 310. IaaS provides virtualized infrastructure resources.

Trick

I = Infrastructure

P = Platform

S = Software

34. Database

- 311. DBMS stands for **Database Management System**.
- 312. DBMS is software used to create, manage and access databases.
- 313. RDBMS stands for **Relational Database Management System**.
- 314. Relational databases organize data into tables/relations.
- 315. A row is commonly called a **record/tuple**.
- 316. A column is commonly called a **field/attribute**.
- 317. A primary key uniquely identifies records in a table.
- 318. A foreign key establishes a relationship/reference between tables.
- 319. SQL stands for **Structured Query Language**.
- 320. SELECT retrieves data.
- 321. INSERT adds data.
- 322. UPDATE modifies data.
- 323. DELETE removes selected records.

- 324. CREATE is a DDL command.
 - 325. ALTER is a DDL command.
 - 326. DROP is a DDL command.
 - 327. TRUNCATE removes table rows while retaining the table structure, subject to DBMS-specific behavior.
 - 328. GRANT and REVOKE are DCL commands.
 - 329. COMMIT and ROLLBACK are TCL commands.
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35. MS WORD

- 330. MS Word is word-processing software.
 - 331. Word is mainly used for creating, editing and formatting documents.
 - 332. **Ctrl + C** = Copy.
 - 333. **Ctrl + X** = Cut.
 - 334. **Ctrl + V** = Paste.
 - 335. **Ctrl + Z** = Undo.
 - 336. **Ctrl + Y** = Redo.
 - 337. **Ctrl + A** = Select All.
 - 338. **Ctrl + S** = Save.
 - 339. **Ctrl + P** = Print.
 - 340. **Ctrl + F** = Find.
 - 341. **Ctrl + H** = Replace.
 - 342. **Ctrl + B** = Bold.
 - 343. **Ctrl + I** = Italic.
 - 344. **Ctrl + U** = Underline.
 - 345. **Ctrl + E** = Center alignment.
 - 346. **Ctrl + L** = Left alignment.
 - 347. **Ctrl + R** = Right alignment.
 - 348. **Ctrl + J** = Justify.
 - 349. Bullets and Numbering are used to create organized lists.
 - 350. Mail Merge is used to create personalized documents for multiple recipients using a common template and data source.
 - 351. Header appears in the upper area of a page.
 - 352. Footer appears in the lower area of a page.
 - 353. Find and Replace can locate text and replace it with another text.
 - 354. Word Wrap automatically moves text to the next line when the current line reaches the margin.
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36. MS EXCEL

- 355. MS Excel is spreadsheet software.
- 356. An Excel file is called a workbook.
- 357. A workbook contains worksheets.
- 358. A worksheet consists of rows and columns.
- 359. The intersection of a row and column is a cell.
- 360. A cell address such as **A1** identifies a cell.
- 361. A formula in Excel generally begins with =.
- 362. **SUM()** adds values.
- 363. **AVERAGE()** calculates the arithmetic mean.
- 364. **MAX()** returns the largest value.
- 365. **MIN()** returns the smallest value.

- 366. COUNT () counts cells containing numbers.
 - 367. COUNTA () counts non-empty cells.
 - 368. IF () performs logical testing.
 - 369. COUNTIF () counts cells satisfying a criterion.
 - 370. SUMIF () sums values satisfying a criterion.
 - 371. Relative reference example = **A1**.
 - 372. Absolute reference example = **\$A\$1**.
 - 373. Mixed reference examples = **\$A1** and **A\$1**.
 - 374. A chart presents spreadsheet data graphically.
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37. PowerPoint

- 375. MS PowerPoint is presentation software.
 - 376. A PowerPoint presentation consists of slides.
 - 377. Slide Transition is an effect between slides.
 - 378. Animation is an effect applied to an object/text on a slide.
 - 379. Slide Master controls common formatting/layout elements across slides.
 - 380. Speaker Notes help the presenter keep additional notes.
 - 381. Slide Show displays the presentation to an audience.
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38. File Extensions

- 382. .docx = Microsoft Word document.
 - 383. .xlsx = Microsoft Excel workbook.
 - 384. .pptx = Microsoft PowerPoint presentation.
 - 385. .pdf = Portable Document Format.
 - 386. .txt = Plain text file.
 - 387. .csv = Comma-Separated Values.
 - 388. .jpg/.jpeg = Image format.
 - 389. .png = Image format supporting transparency.
 - 390. .zip = Compressed archive.
 - 391. .exe = Executable file in Windows environments.
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39. Data Processing

- 392. Data processing converts raw data into useful information.
 - 393. The basic cycle is **Input** → **Processing** → **Output**.
 - 394. Storage may be considered an additional stage in information processing.
 - 395. Batch processing handles data in groups/batches.
 - 396. Real-time processing processes data with very little delay, often under timing constraints.
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40. Artificial Intelligence / Modern Technology

- 397. AI stands for Artificial Intelligence.

- 398. AI aims to enable machines to perform tasks requiring aspects of human-like intelligence.
 - 399. ML stands for Machine Learning.
 - 400. Machine Learning allows systems to learn patterns from data.
 - 401. Deep Learning is a subset of machine learning using multi-layer neural networks.
 - 402. NLP stands for Natural Language Processing.
 - 403. NLP deals with processing and understanding human language.
 - 404. IoT stands for Internet of Things.
 - 405. IoT connects physical devices/sensors to networks for data exchange.
 - 406. Big Data refers to very large, complex and rapidly generated datasets.
 - 407. Blockchain is a distributed ledger technology.
 - 408. Biometrics identifies/authenticates people using biological or behavioral characteristics.
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41. Important Computer Abbreviations

- 409. CPU = Central Processing Unit.
- 410. ALU = Arithmetic Logic Unit.
- 411. CU = Control Unit.
- 412. RAM = Random Access Memory.
- 413. ROM = Read Only Memory.
- 414. SRAM = Static Random Access Memory.
- 415. DRAM = Dynamic Random Access Memory.
- 416. BIOS = Basic Input/Output System.
- 417. UEFI = Unified Extensible Firmware Interface.
- 418. USB = Universal Serial Bus.
- 419. UPS = Uninterruptible Power Supply.
- 420. OCR = Optical Character Recognition.
- 421. OMR = Optical Mark Recognition.
- 422. MICR = Magnetic Ink Character Recognition.
- 423. BCD = Binary Coded Decimal.
- 424. ASCII = American Standard Code for Information Interchange.
- 425. URL = Uniform Resource Locator.
- 426. WWW = World Wide Web.
- 427. HTTP = HyperText Transfer Protocol.
- 428. HTTPS = HyperText Transfer Protocol Secure.
- 429. FTP = File Transfer Protocol.
- 430. SMTP = Simple Mail Transfer Protocol.
- 431. POP = Post Office Protocol.
- 432. IMAP = Internet Message Access Protocol.
- 433. DNS = Domain Name System.
- 434. DHCP = Dynamic Host Configuration Protocol.
- 435. TCP = Transmission Control Protocol.
- 436. UDP = User Datagram Protocol.
- 437. IP = Internet Protocol.
- 438. LAN = Local Area Network.
- 439. MAN = Metropolitan Area Network.
- 440. WAN = Wide Area Network.
- 441. PAN = Personal Area Network.
- 442. DBMS = Database Management System.
- 443. RDBMS = Relational Database Management System.
- 444. SQL = Structured Query Language.
- 445. PDF = Portable Document Format.
- 446. AI = Artificial Intelligence.
- 447. ML = Machine Learning.

- 448. IoT = Internet of Things.
 - 449. NLP = Natural Language Processing.
 - 450. VPN = Virtual Private Network.
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43. AHC RO/ARO — HIGH PRIORITY TRAPS

- 451. RAM is volatile; ROM is non-volatile.
 - 452. Cache is faster than RAM.
 - 453. Registers are faster than cache.
 - 454. OCR reads characters; OMR reads marks; MICR reads magnetic-ink characters.
 - 455. Browser is not a search engine.
 - 456. Internet is not the same as [WWW](http://www.).
 - 457. Router connects networks; switch mainly connects devices within a LAN.
 - 458. MAC address is not the same as IP address.
 - 459. SMTP is mainly for sending email.
 - 460. POP3/IMAP are used for email retrieval/access.
 - 461. HTTP and HTTPS are different.
 - 462. Virus and worm are different.
 - 463. Compiler, interpreter and assembler perform different translation functions.
 - 464. Primary key uniquely identifies a record.
 - 465. Foreign key establishes a reference/relationship between tables.
 - 466. Transition affects movement between slides; animation affects slide objects.
 - 467. Hardware is physical; software is logical/instructional.
 - 468. Data is raw; information is processed/meaningful.
 - 469. Impact printer physically strikes; non-impact printer does not.
 - 470. IPv4 = 32-bit; IPv6 = 128-bit.
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44. MUST-MEMORIZE SHORTCUTS

- Ctrl+A** → Select All
- Ctrl+B** → Bold
- Ctrl+C** → Copy
- Ctrl+E** → Center
- Ctrl+F** → Find
- Ctrl+H** → Replace
- Ctrl+I** → Italic
- Ctrl+J** → Justify
- Ctrl+K** → Hyperlink
- Ctrl+L** → Left Align
- Ctrl+N** → New
- Ctrl+O** → Open
- Ctrl+P** → Print
- Ctrl+R** → Right Align
- Ctrl+S** → Save
- Ctrl+U** → Underline
- Ctrl+V** → Paste
- Ctrl+X** → Cut
- Ctrl+Y** → Redo
- Ctrl+Z** → Undo

45. LAST-MINUTE 30 FACTS

1. CPU → Processing.
2. ALU → Arithmetic + Logic.
3. CU → Control.
4. RAM → Volatile.
5. ROM → Non-volatile.
6. Cache → High-speed memory.
7. SSD → Solid-state storage.
8. Binary → Base 2.
9. Octal → Base 8.
10. Decimal → Base 10.
11. Hexadecimal → Base 16.
12. BCD → Binary Coded Decimal.
13. OCR → Characters.
14. OMR → Marks.
15. MICR → Bank cheques.
16. GUI → Graphical interface.
17. CLI → Command interface.
18. LAN → Local.
19. MAN → Metropolitan.
20. WAN → Wide.
21. Router → Networks.
22. Switch → LAN/device connectivity.
23. DNS → Domain → IP resolution.
24. DHCP → Automatic network configuration.
25. SMTP → Send email.
26. POP3/IMAP → Retrieve/access email.
27. HTTP → Web protocol.
28. HTTPS → Secure web communication.
29. Firewall → Network traffic security/filtering.
30. GIGO → Garbage In, Garbage Out.